CS-477 COMPUTER VISION PROJECT
MIDTERM PROJECT EVALUATION — STAGE 1

SEECS CV Face-Off 2026 | Week 8 | 30 Marks

1. Purpose and Evaluation Framework

Stage 1 (Midterm, Week 8) assesses the independently demonstrated technical progress of each subgroup, the team's GitHub management discipline, and readiness of the Jetson Orin Nano 8GB platform. Stage 2 (Post-Midterm/Final) assesses remaining subgroup development and overall integrated system performance.

The Semester Project Guidebook's overall grade structure remains unchanged: 60 marks for subgroup performance and 40 marks for overall integrated project performance, plus the separately defined Face-Off bonus. Stage 1 contributes 30 of the 60 subgroup marks. Stage 2 will account for the remaining 30 subgroup marks and 40 overall-project marks; its detailed assessment will be issued separately.

2. Stage 1 Assessment Rubric (30 Marks)

Component	Marks	Demonstrated evidence
A. Subgroup Algorithm Selection & Working Demonstration	15	Each subgroup demonstrates a working module or reproducible prototype on representative input; explains the selected primary approach, at least one backup/alternative, interface I/O and preliminary evidence supporting the selection.
B. GitHub Repository & Project Management	10	Team repository and project board operational; subgroup folders/branches and commits; module READMEs and run instructions; assigned tasks, meeting notes, status and blockers documented.
C. Jetson Orin Nano & Camera Setup Readiness	5	Team demonstrates or documents bootable platform, software/environment verification and camera capture readiness; SG-6 leads the common setup, supported by the other subgroups.

3. Detailed Marking Guidance

A. Subgroup Algorithm Selection & Working Demonstration — 15

Criterion	Marks
Working prototype: run live or from saved inputs, with observable, correct module output	7
Primary algorithm selection and backup/alternative, with comparative rationale/evidence	4
V1 input/output contract, reproducibility and explanation of the module's role	2
Preliminary testing/results and known limitations or failure cases	2

A working prototype may run on a development PC at this stage. For software modules, standardized test data or mock upstream inputs are acceptable when a neighboring module is unavailable. The demonstration must clearly show actual module functionality rather than slides alone. The backup approach need not be fully implemented if its feasibility and selection rationale are documented.

B. GitHub Repository & Project Management — 10

Criterion	Marks
One organized shared repository, appropriate module structure, meaningful subgroup commits	3
Project board with owners, statuses, progress and blockers kept current	3
Module README, dependencies, run instructions and evidence links	2
Documented coordination: group meeting record, decisions and interface/dependency tracking	2

This is a team-wide infrastructure assessment. Evidence should identify contributions of each subgroup; marks may be differentiated if an individual subgroup has not maintained its assigned code and documentation.

C. Jetson Orin Nano & Camera Setup Readiness — 5

Criterion	Marks
Jetson Orin Nano 8GB boots and operating environment/storage configuration is identified	2
Core development stack readiness documented (Python, Git, OpenCV; CUDA/PyTorch where required)	1
Camera connection or tested video-acquisition path demonstrated; outstanding blockers and next actions recorded	2

This criterion measures platform readiness, not completion of Jetson deployment for every algorithm. SG-6/Jetson Warriors may lead the setup demonstration, but all subgroups should understand the deployment dependencies. USB or CSI camera is acceptable; a prerecorded-video fallback may demonstrate software readiness but does not by itself prove live camera integration.

4. Evaluation Procedure and Accountability

- Assessment is conducted subgroup-wise for **Component A.** Each subgroup presents its module, executes a working demonstration and answers technical questions.
- **Components B and C** use shared team evidence, checked once per main project team where practical, with subgroup-specific ownership verified.
- Either student in a two-member subgroup may be asked to explain the algorithm, interface, code, test results or contribution. Individual marks may differ if evidence warrants.
- Partial marks are awarded for verifiable completed work. An unavailable neighboring module must not automatically invalidate a subgroup that demonstrates its own correct interface behavior using agreed test inputs.
- All demonstrations must be reproducible from the submitted code and documented dependencies; unsupported claims or screenshots alone do not substitute for working evidence.

5. Scope Boundary and Next Stage

Stage 1 does not require a fully integrated camera-to-decision system on Jetson. The Guidebook places first multi-module integration at Lab 8, individual Jetson module deployment at Lab 9, and the complete embedded pipeline at Lab 11. Teams should be able to identify interface blockers, and provide a clear integration plan; Stage 2 will assess full deployment, validation and end-to-end performance.

Stage 2 (70 marks total): remaining subgroup performance 30 marks + overall integrated project performance 40 marks. The established Face-Off bonus remains subject to the original Guidebook rules.

6. Assessor Record

Assessment component	Maximum	Awarded
A. Subgroup algorithm & demo	15	
B. GitHub & project management	10	
C. Jetson & camera readiness	5	
TOTAL	30	

Project Team: _____ Subgroup: _____ Students: _____

Evaluator remarks / blockers / follow-up: _____